

# Business Case Study

## LoanTap : Logistic Regression

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### Colab Link:

[https://drive.google.com/file/d/1Mw\\_fXrW-O912YNYgrZlqbD79cgfP4zT5/view?usp=sharing](https://drive.google.com/file/d/1Mw_fXrW-O912YNYgrZlqbD79cgfP4zT5/view?usp=sharing)

### Context

LoanTap is an online platform committed to delivering customized loan products to millennials. They innovate in an otherwise dull loan segment, to deliver instant, flexible loans on consumer-friendly terms to salaried professionals and businessmen.

The data science team at LoanTap is building an underwriting layer to determine the creditworthiness of MSMEs as well as individuals.

LoanTap deploys formal credit to salaried individuals and businesses 4 main financial instruments:

1. Personal Loan
2. EMI Free Loan
3. Personal Overdraft
4. Advance Salary Loan

This case study will focus on the underwriting process behind Personal Loan only

### Problem Statement:

Given a set of attributes for an Individual, determine if a credit line should be extended to them. If so, what should the repayment terms be in business recommendations?

### Dataset

Dataset: [LoanTapData.csv](#)

### Data dictionary:

1. loan\_amnt : The listed amount of the loan applied for by the borrower. If at some point in time, the credit department reduces the loan amount, then it will be reflected in this value.
2. term : The number of payments on the loan. Values are in months and can be either 36 or 60.
3. int\_rate : Interest Rate on the loan
4. installment : The monthly payment owed by the borrower if the loan originates.
5. grade : LoanTap assigned loan grade
6. sub\_grade : LoanTap assigned loan subgrade
7. emp\_title : The job title supplied by the Borrower when applying for the loan.\*
8. emp\_length : Employment length in years. Possible values are between 0 and 10 where 0 means less than one year and 10 means ten or more years.
9. home\_ownership : The home ownership status provided by the borrower during registration or obtained from the credit report.
10. annual\_inc : The self-reported annual income provided by the borrower during registration.
11. verification\_status : Indicates if income was verified by LoanTap, not verified, or if the income source was verified
12. issue\_d : The month which the loan was funded
13. loan\_status : Current status of the loan - Target Variable
14. purpose : A category provided by the borrower for the loan request.
15. title : The loan title provided by the borrower

16. **dti** : A ratio calculated using the borrower's total monthly debt payments on the total debt obligations, excluding mortgage and the requested LoanTap loan, divided by the borrower's self-reported monthly income.
17. **earliest\_cr\_line** : The month the borrower's earliest reported credit line was opened
18. **open\_acc** : The number of open credit lines in the borrower's credit file.
19. **pub\_rec** : Number of derogatory public records
20. **revol\_bal** : Total credit revolving balance
21. **revol\_util** : Revolving line utilization rate, or the amount of credit the borrower is using relative to all available revolving credit.
22. **total\_acc** : The total number of credit lines currently in the borrower's credit file
23. **initial\_list\_status** : The initial listing status of the loan. Possible values are – W, F
24. **application\_type** : Indicates whether the loan is an individual application or a joint application with two co-borrowers
25. **mort\_acc** : Number of mortgage accounts.
26. **pub\_rec\_bankruptcies** : Number of public record bankruptcies
27. **Address**: Address of the individual

**Concept Used:**

1. Exploratory Data Analysis
2. Feature Engineering
3. Logistic Regression
4. Precision Vs Recall Tradeoff

## **LoanTap : Logistic Regression**

### **Important Features**

Based on the EDA and published analyses of this particular dataset, important relationships include:

#### **Interest rate**

Higher interest rates are generally associated with higher risk.

#### **DTI**

Higher debt-to-income ratio indicates greater existing debt burden.

#### **Revolving utilization**

High utilization can indicate greater credit stress.

#### **Loan amount**

Larger loans can carry greater exposure.

#### **Term**

The 36-month loans are more likely to be fully paid than the longer-term loans in this dataset.

#### **Grade**

Loan grade is highly informative because it is itself an underwriting/risk assessment variable.

**Questionnaire (Answers should present in the text editor along with insights):**

1. What percentage of customers have fully paid their Loan Amount?
2. Comment about the correlation between Loan Amount and Installment features.
3. The majority of people have home ownership as \_\_\_\_\_.
4. People with grades 'A' are more likely to fully pay their loan. (T/F)
5. Name the top 2 afforded job titles.
6. Thinking from a bank's perspective, which metric should our primary focus be on.
  1. ROC AUC
  2. Precision
  3. Recall
  4. F1 Score
7. How does the gap in precision and recall affect the bank?
8. Which were the features that heavily affected the outcome?
9. Will the results be affected by geographical location? (Yes/No)

**Q1. What percentage of customers have fully paid their loan?**

**Approximately 80.25%.**

Approximately:

Fully Paid = 80.25%  
Charged Off = 19.75%

**Q2. Comment about correlation between Loan Amount and Installment.**

**There is a strong positive correlation between `loan_amnt` and `installment`.**

As loan amount increases, the monthly installment generally increases.

Because of this strong relationship, the two variables create multicollinearity. A published analysis of this dataset found VIF values of approximately **11.58 for loan amount and 11.04 for installment**, and removed `installment` from the final model.

**Q3. Majority of people have home ownership as \_\_\_\_\_.**

**MORTGAGE**

This is also reported in analyses of the dataset.

**Q4. People with grades 'A' are more likely to fully pay their loan. T/F?**

**True**

Grade A represents the lower-risk end of LoanTap's grading system, and borrowers in this group show stronger repayment performance.

**Q5. Name the top 2 afforded job titles.**

**Teacher and Manager**

These are reported as the two job titles receiving the highest number of loans in one analysis of this dataset.

One important distinction:

"Top 2 job titles" means **most frequently occurring job titles**, not necessarily the two highest-income occupations.

#### **Q6. Which metric should the bank primarily focus on?**

If the choices are:

ROC AUC  
Precision  
Recall  
F1 Score

For **default detection**, my primary choice would be:

#### **Recall**

because we want to detect as many actual defaulters as possible.

But in a real bank:

Don't optimize recall blindly. Use a **cost-sensitive threshold** because false positives also have an opportunity cost.

#### **Q7. How does the precision-recall gap affect the bank?**

##### **High precision + low recall**

The bank is conservative about labeling customers as risky.

Result:

Fewer good customers rejected  
BUT  
More actual defaulters missed

##### **Low precision + high recall**

The bank catches more defaulters.

But:

More good customers incorrectly classified as risky

Therefore the bank needs a **balanced operating point** based on business costs.

#### **Q8. Which features heavily affected the outcome?**

The answer can be obtained from:

```
coef_df.sort_values(  
    "Abs_Coefficient",  
    ascending=False  
)head(15)
```

|    | Feature        | Coefficient | Abs_Coefficient | Odds_Ratio |
|----|----------------|-------------|-----------------|------------|
| 54 | zip_code_93700 | 3.114832    | 3.114832        | 22.529649  |
| 47 | zip_code_11650 | 3.086254    | 3.086254        | 21.894902  |
| 53 | zip_code_86630 | 3.047834    | 3.047834        | 21.069661  |
| 52 | zip_code_70466 | 2.797493    | 2.797493        | 16.403465  |
| 50 | zip_code_30723 | 2.784776    | 2.784776        | 16.196193  |
| 51 | zip_code_48052 | 2.781634    | 2.781634        | 16.145385  |
| 48 | zip_code_22690 | 2.780855    | 2.780855        | 16.132805  |
| 49 | zip_code_29597 | -2.574756   | 2.574756        | 0.076172   |
| 46 | zip_code_05113 | -2.570354   | 2.570354        | 0.076508   |
| 6  | dti            | 0.490108    | 0.490108        | 1.632493   |
| 19 | grade_D        | 0.475037    | 0.475037        | 1.608074   |
| 18 | grade_C        | 0.448607    | 0.448607        | 1.566129   |
| 20 | grade_E        | 0.400598    | 0.400598        | 1.492717   |
| 21 | grade_F        | 0.263786    | 0.263786        | 1.301849   |
| 17 | grade_B        | 0.250885    | 0.250885        | 1.285163   |

#### Q9. Will the results be affected by geographical location?

Yes.

Geography can affect:

- income levels
- employment opportunities
- cost of living
- housing conditions
- economic conditions
- local default rates

However, there is a major modeling concern:

Geographic variables can act as **proxies for sensitive characteristics** and can introduce fairness/bias problems.

Therefore, I would **not recommend blindly using full address information**.

If ZIP code is used, it should undergo fairness, stability, and regulatory review.

# Business Recommendation

Instead of saying:

"Approve if probability  $< 0.5$ , reject otherwise."

I would recommend a **three-zone underwriting strategy**.

Low risk

$P(\text{default}) < 0.20$

→ Approve

Medium risk

$0.20 \leq P(\text{default}) < 0.50$

→ Manual review / additional documentation

High risk

$P(\text{default}) \geq 0.50$

→ Reject or offer a smaller loan / shorter term / higher risk-adjusted pricing, subject to lending and regulatory constraints.

**These thresholds are illustrative, not discovered from the dataset.** In a real system, they should be chosen using expected financial loss and business constraints.

## Final Business Recommendation

The final recommendation should be:

LoanTap should use Logistic Regression as an interpretable first-level underwriting model to estimate the probability of default. The model should not rely on accuracy alone because the target is imbalanced. Recall for the Charged Off class should be closely monitored to reduce missed defaulters, while precision should also be considered to avoid unnecessarily rejecting creditworthy borrowers. The probability threshold should be optimized according to the relative financial cost of false positives and false negatives. High-risk applicants can be rejected or offered lower exposure, while medium-risk applicants can undergo additional verification. The model should be periodically monitored for performance, drift, and geographical/fairness-related bias.